

COLLECTION · BUYERS

Clay soils, pyrite and French drains

The Québec-specific field checks — buying with your eyes open

READ BEFORE YOU BEGIN

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The Québec-specific field checks — buying with your eyes open

Some building risks are specific to Quebec and the Montréal region: clay soils, pyrite, an aging French drain, buried tanks, vermiculite. Well identified, they can be managed; ignored, they cost a lot.

This guide teaches you to spot the signals, ask the right questions and target the professional checks before buying.

This content is informational: diagnoses must be made by professionals (inspector, engineer, expert). Each chapter ends with a lined “My notes” page.

The Quebec building stock has its particularities: clay soils, pyrite and pyrrhotite, French drains, buried tanks. Well known, these issues can be managed; ignored, they cost a lot. This guide teaches you to spot them and protect yourself before buying.

Goal: turn scary words into concrete check points for your inspection.

CHAPTER

Clay soils and foundation movement

Understanding why clay moves and what it implies.

A large part of the Montréal, Laval and North Shore region sits on clay soils. Clay swells with water and shrinks in dry periods, which can cause foundation movement, cracks and uneven floors — especially during droughts or with large trees nearby that dry out the soil.

- Signals: stair-step cracks, sticking doors/windows, leaning floors.
- Aggravating factors: mature trees near the foundations, poor drainage, droughts.
- When in doubt: have it assessed by a professional (sometimes an engineer).

Clay swells with water and shrinks in dry periods, which can move foundations (cracks, leaning floors). It's not inevitable: good water management around the house (slopes, gutters, stable watering of the foundations in a drought) strongly limits movement.

EMILIE'S TIP

Not all cracks are serious, but some are. In a clay area, have significant cracks assessed by a pro before buying: it's the difference between a cosmetic detail and a major bill.

CHAPTER

Pyrite and pyrrhotite

Recognizing these mineral risks by area.

Pyrite can be present in the backfill under the basement slabs of certain areas: as it swells, it lifts and cracks the slab. Pyrrhotite, for its part, affects the foundation concrete of certain regions (mostly outside Greater Montréal, but worth knowing) and can degrade it.

- Pyrite signals: lifting and cracking of the basement slab, pressure on the partitions.
- Check: a swelling test of the backfill (petrography) by a specialized lab.
- Some areas are more at risk: inquire locally.

Pyrite (under the slabs) and pyrrhotite (in the concrete of certain regions) can, by reacting, crack and lift the concrete. The risk varies by area and era of construction. Specialized tests exist: when in doubt, do them before buying.

EMILIE'S TIP

If a basement slab shows cracks and lifting in an area known for pyrite, require a test before buying. The cost of the test is trivial compared with that of a slab to be redone.

CHAPTER

The French drain and water management

Assessing the system that keeps the basement dry.

The French drain, buried at the base of the foundations, drains water from the soil. Over time (often a few decades), it can clog or sag, leading to infiltration in the basement. Replacing it is a costly job (excavation).

- Signals: dampness or infiltration in the basement, efflorescence, odours.
- Good surface drainage: the slope of the lot and gutters carrying water away from the house.
- A camera can inspect the drain's condition when in doubt.

The French drain, buried at the base of the foundations, drains water from the soil; clogged or collapsed, it lets water seep into the basement. Replacing it is expensive. Observe the clues (dampness, stains, odour) and ask for the drain's age and history.

EMILIE'S TIP

The French drain's age is a question to ask systematically on a house over 30 years old. A dry basement today doesn't guarantee a drain in good condition.

CHAPTER

Buried tanks, vermiculite and asbestos

Spotting the other Québec-specific points of vigilance.

- Buried oil tank (old oil heating): a risk of soil contamination and an insurability issue; its presence must be declared and managed.
- Vermiculite (granular insulation in some old attics/walls): may contain asbestos; to be tested before any work.
- Other asbestos-containing materials in older buildings: precautions during renovations.

None of these elements is necessarily a deal-breaker, but each must be identified, assessed and, if needed, costed before the purchase.

Stay alert to other Québec-specific points: a buried oil tank (contamination), vermiculite that may contain asbestos, and old at-risk materials. These can be managed, but it's better to know them before buying than to discover them while renovating.

EMILIE'S TIP

These points shouldn't make you flee every old house — they should make you ask questions and target the inspection. A well-understood house is one you buy with peace of mind.

CHAPTER

Detecting these risks and protecting yourself

Checking before buying.

These soil and building problems are well managed... when detected in time. Your protections at purchase:

- **A pre-purchase inspection** by a professional attentive to these regional issues.
- **Specialized tests** if needed (pyrite, pyrrhotite, soil).
- **Questions to the seller** and careful reading of their declaration.
- **The history** of work and repairs.

An inspection condition in your promise to purchase gives you time to validate all this.

Your best protection: an inspection by a professional attentive to these regional issues, targeted tests if needed, and an inspection condition in your promise to purchase. A few hundred dollars of expertise can save tens of thousands in repairs.

EMILIE'S TIP

Always add an inspection condition to your offer, especially in areas known for these issues. A few hundred dollars of expertise can save you tens of thousands in repairs.

IN PRACTICE

Action plan

Your next steps, concretely.

This week

1. Find out whether my target areas are at risk (clay, pyrite).
2. List the signals to watch for during visits.

Within 30 days

1. Target the relevant field checks with the inspector.
2. Plan specialized tests (pyrite, drain, vermiculite) if necessary.

Next

1. Factor any identified risk into my price and condition strategy.

To protect myself

1. Add an inspection condition and target the area's issues.
2. Plan specialized tests (pyrite, soil) when in doubt.

REFERENCES

Resources & glossary

- Clay soil: soil that swells and shrinks, potentially moving the foundations.
 - Pyrite: a mineral in the backfill that swells and lifts the basement slab.
 - Pyrrhotite: a mineral that can degrade foundation concrete (certain regions).
 - French drain: a pipe at the base of the foundations draining water from the soil.
 - Vermiculite: old granular insulation that may contain asbestos.
 - Buried tank: an old oil tank, a soil and insurance issue.
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- Clay soil: soil that swells and shrinks with moisture, potentially moving the foundations.
 - Pyrite / pyrrhotite: minerals that can swell or crack concrete.
 - French drain: a perimeter drain carrying water away around the foundations.
 - Buried tank: an old oil tank, a contamination risk.
 - Vermiculite: insulation that may contain asbestos.

GOING FURTHER

Quiz — test your knowledge

1. Clay soils can cause:

- A) Nothing
- B) Foundation movement and cracks
- C) Better insulation
- D) Energy savings

2. An aggravating factor in clay soil is:

- A) A small lawn
- B) Large trees near the foundations
- C) A new roof
- D) A new door

3. Pyrite mostly affects:

- A) The roof
- B) The backfill under the basement slab
- C) The windows
- D) The electrical system

4. To check for pyrite, you do:

- A) A photo
- B) A swelling test (petrography) in a lab
- C) Nothing
- D) A water test

5. The French drain serves to:

- A) Heat the house
- B) Drain water from the soil at the base of the foundations
- C) Insulate the roof
- D) Supply water

6. The drain's age is a key question on a house over:

- A) 5 years
- B) 10 years
- C) 30 years
- D) 100 years only

7. Vermiculite may contain:

- A) Gold
- B) Asbestos
- C) Drinking water
- D) Concrete

8. A buried oil tank is mainly an issue of:

- A) Decoration
- B) Soil contamination and insurability
- C) Free heating
- D) Nothing

Answer key

- 1. **B** — Clay swells and shrinks (ch. 1).
- 2. **B** — Trees dry out the soil (ch. 1).
- 3. **B** — It swells and lifts the slab (ch. 2).
- 4. **B** — A specialized analysis of the backfill (ch. 2).
- 5. **B** — It keeps the basement dry (ch. 3).
- 6. **C** — It ages over a few decades (ch. 3).
- 7. **B** — To be tested before any work (ch. 4).
- 8. **B** — An environmental and insurance risk (ch. 4).

Signé PAR Em

*Thank you for reading. May this guide help you
decide with confidence, with all the information
in hand.*

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